

Apple Containers CLI Cheatsheet

Practical command reference for 'container' on macOS.
Inspired by and adapted from the official Apple 'container' repository documentation.

System
Service Management <pre># Starts Apple Containers services container system start # Stops Apple Containers services container system stop</pre>
Info & Help <pre># Shows CLI + API server version details container system version --format table # Lists top-level commands and options container --help</pre>
Service Logs <pre># Displays logs from the container services container system logs # Follows logs in real time container system logs --follow # Shows only the last N log lines/duration container system logs --last 1h</pre>
List Containers <pre># Lists all containers (running + stopped) container ls -a</pre>
Build an Image <pre># Creates a new directory for the Dockerfile and context mkdir web-test cd web-test</pre>
Dockerfile Example <pre>FROM docker.io/python:alpine WORKDIR /content RUN echo 'hello' > index.html CMD ["python3", "-m", "http.server", "80", "--bind", "0.0.0.0"]</pre>
Build <pre># Builds the image from the Dockerfile in the current directory container build --tag web-test --file Dockerfile . # Builds without using the cache container build --tag web-test --no-cache . # Passes a build-time variable container build --tag web-test --build-arg VERSION=1.2 . # Builds for a specific platform container build --tag web-test --platform linux/arm64 . # Builds a specific stage in a multi-stage Dockerfile container build --tag web-test --target builder .</pre>
If -f/--file is omitted, the build looks for Dockerfile first, then falls back to Containerfile.
Image Management <pre># Lists all images container image list # Lists images with verbose / quiet output container image list -v container image list -q # Pulls an image from a registry container image pull docker.io/library/nginx:latest # Pushes an image to a registry container image push myrepo/web-test:latest</pre>

Run & Inspect Containers
Run a Container
<pre># Runs a container and maps host port 8080 to container port 80 container run --name my-web-server --detach --rm --publish 8080:80 web-test # Runs interactively with a shell attached container run -it ubuntu:latest /bin/bash # Sets an environment variable in the container container run -e NODE_ENV=production web-test # Limits CPU and memory usage container run --cpus 2 --memory 1G web-test # Bind mounts a host directory into the container container run -v ./data:/content web-test</pre>
<pre>--detach — runs the container in the background (no terminal attached). --rm — automatically removes the container when it stops. --publish — maps a host port to a container port (host:container). -v/--volume — mounts a host path or named volume into the container.</pre>
Create Without Starting
<pre># Creates a container without starting it (like docker create) container create --name my-web-server web-test</pre>
Execute a Command Inside a Running Container
<pre># Executes a one-off command inside the running container container exec my-web-server ls /content # Opens an interactive shell session inside the container container exec --interactive --tty my-web-server sh # Runs a command detached from the terminal container exec -d my-web-server long-task.sh # Runs a command with a custom working directory and env variable container exec -w /content -e DEBUG=1 my-web-server env</pre>
Run exit to leave the interactive shell session.
Copy Files To/From a Container
<pre># Copies a file from the host into the container container cp ./config.json my-web-server:/etc/app/ # Copies a file from the container to the host container cp my-web-server:/var/log/app.log ./logs/</pre>
Resource Stats
<pre># Displays live resource usage statistics for the running container container stats my-web-server # Displays a single snapshot of resource usage (no continuous stream) container stats --no-stream my-web-server</pre>
Inspect Metadata
<pre># Displays detailed metadata about the container container inspect my-web-server</pre>
View Logs
<pre># Displays the logs from the container's stdout and stderr container logs my-web-server</pre>
Lifecycle Operations
Stop a Running Container
<pre># Stops a running container by name or ID container stop my-web-server</pre>
Stop / Start Cycle
<pre># Runs a detached container named test-stopstart container run --name test-stopstart --detach web-test # Stops the container container stop test-stopstart # Starts it again container start test-stopstart # Deletes the container container delete test-stopstart</pre>

Volume Management	
Create & List	
# Creates a named volume <code>container volume create myvolume</code>	# Creates a volume with a size limit and ordered journaling <code>container volume create --opt size=10g --opt journal=ordered myvolume</code>
# Lists all volumes <code>container volume list</code>	# Lists only volume names <code>container volume list -q</code>
Delete & Prune	
# Deletes a single volume by name <code>container volume delete myvolume</code>	# Deletes multiple volumes at once <code>container volume delete vol1 vol2 vol3</code>
# Deletes all volumes <code>container volume delete --all</code>	# Removes all volumes with no container references <code>container volume prune</code>
Anonymous volumes — created via <code>-v /path</code> without a source (named <code>anon-{uuid}</code>). Unlike Docker, anonymous volumes do not auto-cleanup with <code>--rm</code> . Manual deletion is required.	
Builder Management	
# Shows BuildKit builder container status <code>container builder status</code>	# Stops the BuildKit builder container <code>container builder stop</code>
# Deletes the builder container <code>container builder delete</code>	# Force-deletes the builder even if still running <code>container builder delete --force</code>
Cleanup	
# Deletes a container by name or ID <code>container delete my-web-server</code>	# Removes all stopped containers to reclaim disk space <code>container prune</code>
# Stops the Apple Containers services <code>container system stop</code>	
Network Management (macOS 26+)	
Available on macOS 26 and later only. Allows creation and management of user-defined container networks.	
Create & List	
# Creates a new user-defined network <code>container network create mynet</code>	# Creates an internal-only network (no external routing) <code>container network create --internal mynet</code>
# Creates a network with a custom subnet <code>container network create --subnet 10.10.0.0/24 mynet</code>	
# Lists all networks <code>container network list</code>	
Delete	
# Deletes a network by name <code>container network delete mynet</code>	
Use a Network	
# Runs a container attached to a custom network <code>container run --network mynet web-test</code>	# Runs a container with a specific MAC address on a network <code>container run --network default,mac=02:42:ac:11:00:02 ubuntu:latest</code>

Create

Creates and boots a container machine from an image

container machine create alpine:3.22

Creates a named machine with custom CPU/memory, without booting

container machine create --name my-machine --cpus 4 --memory 4G --no-boot alpine:3.22

Run Inside a Machine

Opens an interactive shell in the default container machine

container machine run

Runs a command in a named machine

container machine run -n my-machine uname -a

Passes arguments to the command after --

container machine run -n my-machine -- cat /proc/cpuinfo

List & Manage

Lists all container machines (default marked in DEFAULT column)

container machine list

Stops a named machine

container machine stop -n my-machine

Deletes a named machine

container machine delete -n my-machine

Advanced Run Flags

Runs with an init process to reap zombies and forward signals

container run --init ubuntu:latest my-app

Runs with a custom init image for boot customization

container run --init-image local/custom-init:latest ubuntu:latest

Mounts the container's root filesystem as read-only

container run --read-only web-test

Adds a tmpfs mount at a given path

container run --tmpfs /tmp web-test

Sets a custom shared memory size for /dev/shm

container run --shm-size 128M web-test

Enables Rosetta translation for x86 binaries

container run --rosetta web-test

Forwards the host SSH agent socket into the container

container run --ssh web-test

Exposes virtualization capabilities to the container

container run --virtualization web-test

Sets a resource ulimit (soft:hard)

container run --ulimit nofile=1024:2048 web-test

System: Kernel, Properties & DNS

Kernel Management

Downloads and installs the recommended default kernel

container system kernel set --recommended

Installs a custom kernel binary

container system kernel set --binary ./vmlinux --arch arm64

Installs a kernel from a tar archive with digest verification

container system kernel set --tar https://example.com/kernel.tar --digest sha256:abcd1234

System Properties

Lists all system properties (default TOML format)

container system property list

Lists properties as JSON for scripting

container system property list --format json

DNS Domains

Lists configured container DNS domains

container system dns list

Lists DNS domains as JSON, names only

container system dns list --format json -q